Matheus Hoffmann

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👋 me

sr. software engineer & jr. researcher with **8 years** of experience building scalable (and occasionally clever) solutions mainly for payment processing & fraud-fighting. my research topics are wave modelling, game theory and self-improving apps. i'm currently writing and doing research for **marola-dev** (volunteer), and i'm open to commercial offers.

🌊 marola

i'm actively building **marola** in the open: a citizen-science **ocean intelligence layer** for the brazilian coast. it answers *"can i swim tomorrow, and when?"* for every beach around florianópolis, rio de janeiro and salvador: the best hour from swell, wind, tide, uv and the official bathing-water bulletins, with the *why* in plain language, live at **marola.dev**.

- **water quality first** - every official sampling point (ima/sc, inea, inema) is on the map, coloured by its last bulletin; unfit water zeroes the score in deterministic scala 3 (kyo) code, and the llm explains but never overrules
- **the sea in front of you** - tide, swell, wind, uv, jellyfish and whale odds per beach, a sourced *did you know?* about the local sea, and *ask the ocean*: rag with citations on ollama, an mcp tool server and its own small model, marola-sea, no account or api key
- **designed in public** - every non-trivial change starts as a public design doc (MIPs); next up: wind, wave, satellite and sea-heat map layers, *ressaca* (storm surf) warnings, wave-model ensembles with WAVEWATCH III and an open data lake of beaches and water quality
- **one umbrella, many repos** - marola holds no code, only the plans, the shared rules for people and coding agents, and every other repo as a submodule; each repo builds and releases alone. every claude code skill and agent marola runs is listed in marola-skills and marola-agents

🌊 marola.dev · FOSS marola-dev · docs docs.marola.dev · license MIT · DOI 10.5281/zenodo.23224155

civic-tech · disaster-risk-management · forecasting · meteorology · non-profit · oceanography · water-quality

> 🔓 contributors welcome, humans and ai agents alike — no code needed: use the map and tell us where it's wrong, share local sea knowledge, or open an issue or a PR at **marola-dev/marola**. or mail me: **mhoffmannfs[at]gmail.com**

⚡ research — ww3-gpu

my undergraduate project at ufrj (escola politécnica), co-advised at labeco/ufsc: **ww3 gpu lab**, an open lab and a book in progress on agentic coding and agentic research, around one problem: moving the expensive kernels of noaa's **WAVEWATCH III** wave model to gpus *without changing the answer*. a nix-pinned fortran/mpi/netcdf toolchain from **picasso**, my nix labs, builds ww3 and runs a regression test in one command. the dia nonlinear-interaction kernel (`W3SNL1`), ported to c++/**kokkos**, matches the fortran bit for bit on serial, openmp and cuda, and runs 1,000 sea points in 0.047 ms on an rtx 4090 against 24.96 ms serial. its 17 lessons, from the wave spectrum to porting kernels with coding agents, are the book's chapters, built into a pdf and a website on every merge.

research ww3-gpu · DOI 10.5281/zenodo.23221351 · nix labs picasso · DOI 10.5281/zenodo.23311208

📚 closed-source

a non-profit essay, where i'm studying licenses: **closed source: the god that failed**, a position paper on reverse engineering and the ethics of citation in the age of coding agents. a v0 preprint, not peer reviewed.

site h0ffmann.github.io/closed-source · essay closed-source · DOI 10.5281/zenodo.23333656

non-profit · studying licenses

💼 exp-highlights

LIOC (UFRJ) · Stone · Broad (YC W21) · Elemeno AI · Signifyd (US) · TXP (UK, formerly Vigil)

references available on request (ways of working)

🏅 certs

aws certified 4x (Cloud Practitioner, Machine Learning Specialty, Machine Learning Engineer Associate, Data Engineer Associate) · **padi certified 3x** (Advanced Open Water Diver, Enriched Air Nitrox Diver, Rescue Diver)

🤿 misc

Strava open water swimming